

Validated Synthetic Patient Generation for Small Longitudinal Cohorts: Coagulation Dynamics Across Pregnancy

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Abstract

Small longitudinal clinical cohorts, common in maternal health, rare diseases, and early-phase trials, limit computational modeling: too few patients to train reliable models, yet too costly and slow to expand through additional enrollment. We present multiplicity-weighted Stochastic Attention (SA), a generative framework based on modern Hopfield network theory that addresses this gap. SA embeds real patient profiles as memory patterns in a continuous energy landscape and generates novel synthetic patients via Langevin dynamics that interpolate between stored patterns while preserving the geometry of the original cohort. Per-pattern multiplicity weights enable targeted amplification of rare clinical subgroups at inference time without retraining. We applied SA to a longitudinal coagulation dataset from 23 pregnant patients spanning 72 biochemical features across 3 visits (pre-pregnancy baseline, first trimester, and third trimester), including rare subgroups such as polycystic ovary syndrome and preeclampsia. Synthetic patients generated by SA were statistically, structurally, and mechanistically indistinguishable from their real counterparts across multiple independent validation tests, including an ordinary differential equation model of the coagulation cascade. A downstream utility test further showed that a mechanistic model calibrated entirely on synthetic patients predicted held-out real patient outcomes as well as one calibrated on real data. These results demonstrate that geometry-preserving generative methods can produce clinically useful synthetic cohorts from very small longitudinal datasets, enabling data-augmented modeling in small-cohort settings.

Keywords: synthetic data generation, stochastic attention, modern Hopfield networks, multiplicity weighting, coagulation, pregnancy, longitudinal data, mechanistic validation

1 Introduction

In 2021, the United States maternal mortality rate reached 32.9 deaths per 100,000 live births, the highest level since 1965, with Black mothers dying at nearly three times the rate of White mothers [Toy, 2023, Harris, 2023]. Thrombotic embolisms and hemorrhage together accounted for more than 22% of these deaths [Troost et al., 2022], highlighting the role of the coagulation system

in maternal outcomes. Blood coagulation is a complex enzymatic cascade in which a small tissue factor (TF) stimulus triggers a series of serine protease activation reactions that ultimately produce thrombin, the central effector of hemostasis [Mann et al., 2011, Butenas et al., 2009]. Thrombin cleaves fibrinogen into fibrin, activates platelets, and amplifies its own production through positive feedback via Factors V, VIII, and XI, while simultaneously initiating its own shutdown through the thrombomodulin–protein C anticoagulant pathway [Hockin et al., 2002]. Pregnancy alters this hemostatic balance, shifting toward a hypercoagulable state that prepares for delivery [Hellgren, 2003, Grouzi et al., 2022, Bremme, 2003]: procoagulant factors including fibrinogen, Factor VIII, and von Willebrand factor increase substantially across gestation, while natural anticoagulants such as antithrombin and protein S decline. Conditions such as polycystic ovary syndrome (PCOS), a hormonal and metabolic disorder affecting 6–12% of reproductive-age women [Azziz et al., 2004] that has been associated with secondary hemostatic changes including elevated Factor VIII and impaired fibrinolysis, and preeclampsia (PE), a hypertensive disorder characterized by endothelial dysfunction, platelet activation, and altered thrombin generation that complicates 2–8% of pregnancies worldwide [Abalos et al., 2014], further shift this balance in ways that are not fully understood. Mathematical models of the coagulation cascade, from the original Hockin–Mann stoichiometric framework [Hockin et al., 2002] through extensions incorporating the thrombomodulin–protein C pathway [Bravo et al., 2012] to the Brummel-Ziedins 2012 (BZ2012) model with detailed activated protein C (APC)-mediated Factor Va inactivation kinetics [Brummel-Ziedins et al., 2012], have provided mechanistic insight into patient-specific thrombin generation [Luan et al., 2007, 2010], but applying these models to pregnancy cohorts requires longitudinal datasets capturing the full complement of coagulation factors, inhibitors, and thrombin generation assay (TGA) measurements across gestation, the type of data that is rare and expensive to collect.

The core barrier is sample size. Longitudinal pregnancy coagulation studies typically enroll tens of patients with complete follow-up across multiple trimesters, yielding datasets where the number of features p (coagulation factors, TGA parameters, viscoelastic measurements) exceeds the number of patients n . Rare complications compound this problem: small longitudinal cohorts inevitably contain only a handful of patients with any given condition, far too few for condition-specific statistical analysis. This $n < p$ regime limits conventional approaches because covariance matrices are rank-deficient, cross-validation is unreliable, and overfitting is nearly inevitable. Synthetic data generation addresses this by augmenting small real datasets with statistically faithful synthetic records, enabling downstream analyses (mechanistic modeling, clinical comparisons, hypothesis generation for rare complications) that would otherwise be infeasible [Gonzales et al., 2023, Murtaza et al., 2024]. However, existing methods face limitations in the small-cohort longitudinal setting. Sampling from a fitted multivariate normal (MVN) distribution is singular when $n < p$ and must be regularized [Ledoit and Wolf, 2004], introducing bias that distorts the joint distribution. MVN also assumes Gaussian marginals and linear correlations, and provides no mechanism for conditional generation: it cannot amplify a specific subpopulation while preserving its signatures. More expressive models such as GANs and VAEs [Goodfellow et al., 2014, Kingma and Welling, 2014, Xu et al., 2019], including recent longitudinal extensions [Murtaza et al., 2024], require substantially larger training sets and are prone to mode collapse at small n (we confirm this empirically for CTGAN at $n=23$ in this work). A generative method is needed that can operate directly on the geometry of a small dataset without estimating a full parametric distribution.

Stochastic Attention (SA) is such a framework, based on modern Hopfield network theory [Ramsauer et al., 2021], in which stored patient profiles serve as memory patterns in a continuous energy landscape and Langevin dynamics generates novel samples that interpolate between them via the attention mechanism. We recently introduced a multiplicity-weighted extension of the Hopfield energy that adds per-pattern weights to the log-sum-exp term, enabling continuous interpolation

between unconditioned generation and targeted subpopulation amplification at inference time without retraining [Varner, 2026a]. SA operates in a principal component analysis (PCA)-reduced linear subspace of the data rather than fitting a parametric distribution, making it well-suited to the $n < p$ regime. In prior work, we generated synthetic patient populations using MVN distributions fitted independently to each clinical visit [Bravo et al., 2023]; while this captured per-visit statistics, it assumed Gaussian marginals and generated each visit independently rather than as part of a coherent longitudinal trajectory.

In this study, we extended multiplicity-weighted SA to generate complete longitudinal patient profiles from a very small pregnancy coagulation cohort ($K=23$ patients, 72 features per visit, 3 visits) and validated the resulting synthetic patients at four levels: marginal plausibility, cross-visit covariance preservation, conditional amplification of rare subgroups (PCOS, $n=3$; Developed PE, $n=5$), and mechanistic consistency with an independent ordinary differential equation (ODE) model of the coagulation cascade [Brummel-Ziedins et al., 2012]. We developed a pipeline that concatenates each patient’s multi-visit profile into a single vector, applies PCA for dimensionality reduction, and uses multiplicity-weighted Langevin dynamics on the Hopfield energy landscape to generate new patients, with a direction-magnitude decomposition that preserves the natural variance structure of continuous clinical measurements. SA-generated patients reproduced per-feature means with median error of 1.4%, preserved cross-visit correlations that MVN’s rank-deficient covariance could not represent, and maintained condition-specific signatures when amplifying rare subgroups. An independent 58-species ODE model of the coagulation cascade could not distinguish real from synthetic patients, and a mechanistic model calibrated entirely on synthetic patients predicted held-out real patient outcomes as well as one calibrated on real data.

2 Results

We generated $N=100$ synthetic longitudinal patient profiles using SA, chosen to provide approximately $4\times$ amplification over the real cohort while remaining within the regime where SA’s attention mechanism produces diverse, non-degenerate samples, and compared them against the $K=23$ real patients across four validation levels. As a baseline, we also generated $N=100$ patients from a regularized multivariate normal (MVN) distribution fitted to the same data.

2.1 Marginal Plausibility

We first assessed whether SA-generated patients reproduced per-feature summary statistics across all three visits. The median mean relative error (MRE) across all 72 features was 1.4% (95% bootstrap CI: 1.2–1.8%), with the majority of features below 5% (Table 2), indicating that SA captured the central tendencies of the real population. Representative coagulation factors, including Factor II (MRE $< 2\%$), Factor VIII (MRE $< 2\%$), antithrombin (MRE $< 3\%$), and fibrinogen (MRE $< 1\%$), showed strong agreement across all visits. MVN also performed well at this level (median MRE $\approx 2\text{--}3\%$), which was expected since moment-matching is guaranteed by construction. We verified that synthetic patients were not memorized copies of real patients using two complementary metrics. First, the mean novelty score (defined as $1 - \max_k \cos(\hat{\xi}, \hat{\mathbf{m}}_k)$) was 0.50 at $\beta = \beta^*$, indicating that generated samples were on average 50% away from their nearest stored pattern in angular distance (Supplementary Table S1). Second, we computed nearest-neighbor distances in standardized feature space: the median distance from each synthetic patient to their closest real patient was 6.14 (standardized units), compared with 6.70 for the median real-to-real nearest-neighbor distance, a ratio of 0.92 indicating that synthetic patients were approximately as far from real patients as real patients were from each other.

We next examined whether known physiological relationships were preserved in the synthetic cohort. We plotted pairwise scatter plots of coagulation factor levels against thrombin generation parameters across all visits and observed that SA-generated patients occupied the same joint regions as real patients (Fig. 1). The expected inverse relationship between antithrombin and TF-initiated peak thrombin was preserved, as were the positive associations between Factor VIII and peak, prothrombin and endogenous thrombin potential (ETP), and the inverse relationship between antithrombin and ETP. These results indicated that SA captured not only univariate distributions but also the pairwise dependencies in the data.

We also evaluated whether the characteristic pregnancy-driven longitudinal trajectories were reproduced. We computed per-visit means and standard deviations for six key coagulation features and found that synthetic patients tracked the real trajectories closely, with overlapping confidence bands at all three visits (Fig. 2). Fibrinogen, Factor VIII, and von Willebrand factor showed the expected monotonic increases from baseline through the third trimester in both cohorts, while Factor II increased modestly and antithrombin declined then partially recovered, consistent with the known hypercoagulable shift of pregnancy. These patterns confirmed that the concatenated-visit SA pipeline preserved the directionality and magnitude of longitudinal change. The 72 features also included viscoelastic (ROTEM) and fibrinolytic parameters, which were generated as part of the concatenated vector and showed comparable MREs to the coagulation factors (Supplementary Table S5), but are not individually highlighted here because the mechanistic validation uses the BZ2012 thrombin generation model, which does not cover fibrinolysis or clot viscoelasticity.

2.2 Cross-Visit Covariance Structure

Having established that SA reproduces marginal distributions and pairwise relationships, we next asked whether it preserves the higher-order joint structure across visits. We compared the full 216×216 cross-visit correlation matrices for real, SA-generated, and MVN-generated populations. The real data exhibited a characteristic block structure with strong within-visit correlations along the diagonal and structured off-diagonal blocks reflecting cross-visit dependencies; for example, a patient’s Visit 1 Factor X level predicting their Visit 3 Factor X level. We found that SA preserved this block structure, including the off-diagonal cross-visit correlations (Fig. 3). The SA–Real residual matrix showed small, unstructured deviations distributed throughout, whereas the MVN–Real residual was notably smoother, reflecting the regularization-induced shrinkage of off-diagonal correlations toward zero. This difference was most apparent in the off-diagonal blocks, where MVN systematically underestimated cross-visit dependencies that SA retained. We note that Pearson correlation captures linear associations; Spearman rank correlations, used in the mechanistic validation, would additionally capture monotonic nonlinear dependencies.

We examined the eigenvalue spectrum of the sample covariance matrix to understand the structural origin of this difference (Supplementary Fig. S2). The real data had rank 22, reflecting the $K=23$ patient constraint. SA and MVN make different trade-offs in handling this rank deficiency: SA truncates via PCA (zeroing variance beyond component 18), while MVN regularizes via Ledoit–Wolf shrinkage (providing a full-rank estimate by inflating eigenvalues in the 194 null dimensions). The consequence of MVN’s approach is that it introduces variance in dimensions where the data contain no signal, which manifested as increased dispersion in PCA projections: SA-generated patients occupied the same region of PCA space as real patients across all three visits, while MVN-generated patients showed substantially greater scatter, particularly at Visits 2 and 3 (Fig. 4).

2.3 Conditional Generation of Rare Subgroups

The previous analyses used unconditioned SA, generating from the full cohort. A key clinical application, however, is generating patients from specific subpopulations that are too small to study independently. We therefore tested whether SA could amplify rare clinical subpopulations while preserving their condition-specific signatures. We defined three overlapping subgroups by pregnancy outcome and comorbidity: Uncomplicated ($n=18$), PCOS ($n=3$, cross-cutting; 1 patient also in the PE group), and Developed PE ($n=5$). The PCOS and Developed PE groups were too small for any independent statistical analysis. We used SA’s multiplicity-weighted sampling to generate condition-specific cohorts of 100 patients each by upweighting attention on the relevant stored patterns.

We compared condition-specific feature means between real and SA-generated patients across eight coagulation features that exhibited between-condition variation (Fig. 5). SA-generated cohorts preserved the condition-specific patterns: PCOS patients showed elevated Factor VIII and vWF relative to Healthy patients in both real and synthetic data, PE patients showed elevated α_2 -AP and Factor IX, and the between-condition rank ordering was maintained across all eight features. To quantify equivalence, we performed a bootstrap Mann–Whitney test: for each feature and condition, we subsampled the synthetic cohort to match the real sample size (n_{real}), applied a Mann–Whitney U test, and repeated 1,000 times. We reported the fraction of replicates where $p > 0.05$ (i.e., real and synthetic were statistically indistinguishable). Across all 24 feature–condition pairs, 19 (79%) achieved $p > 0.05$ in $\geq 90\%$ of replicates, with a median non-significance fraction of 98.4%. The five features that fell below 90% were distributed across high-variance features (FIX, FVII in the Uncomplicated group; plasminogen in PCOS; FVIII and plasminogen in Developed PE) where the real data exhibited large inter-patient variability. No multiple-testing correction was applied to the 24 simultaneous comparisons; the failing features had consistently low non-significance fractions (68–88%) rather than borderline values, so correction would not change the classification. PCA projections of the conditioned cohorts are provided in Supplementary Figs. S3–S4. These results showed that SA’s multiplicity-weighted interpolation can amplify rare subgroups in a way that preserves their distinguishing clinical features, useful for hypothesis generation and power analysis in small patient populations. MVN has no mechanism for this: fitting a separate MVN to three PCOS patients across 216 features is mathematically impossible (rank 2 covariance), and post-hoc subsetting of unconditional MVN samples does not preserve subgroup-specific structure.

2.4 Mechanistic Consistency

The preceding validations assessed statistical properties of the synthetic data. We next tested whether SA-generated patients produce biologically plausible outputs when their coagulation factor levels are fed through an independent mechanistic model that knows nothing about the SA generation process? We used the Hockin–Mann BZ2012 model [Hockin et al., 2002, Brummel-Ziedins et al., 2012], a system of 58 ordinary differential equations with 64 rate constants that simulates thrombin generation from patient-specific coagulation factor inputs. We calibrated 5 of the 64 rate constants (prothrombinase, intrinsic and extrinsic tenase, protein C activation, and meizothrombin conversion; Supplementary Table S4) on Visit 1 real patients at the population level (not per-patient), held the remaining 59 at published literature values, and ran the calibrated model on all 23 real patients (3 visits $\times$ 2 conditions = 138 simulations, 0 failures) and all 100 synthetic patients (3 visits $\times$ 2 conditions = 600 simulations, 0 failures). For each patient, we computed five ODE-predicted thrombin generation features (lagtime, peak, time-to-peak, maximum rate, and ETP) and compared them against the corresponding values from the patient record. In

these comparisons (Fig. 6, top row), the horizontal axis is the TGA value from the patient record, which exists for both real and synthetic patients, while the vertical axis is the value computed by the BZ2012 model from that patient’s factor inputs. Systematic deviations from the $y=x$ line reflect ODE model bias (e.g., lagtime is underpredicted at $\approx 0.77\times$), not a failure of the synthetic data; the key observation is that real and synthetic patients share the same pattern of model bias, occupying the same cloud with the same systematic offset.

We formalized this observation by computing the ODE-predicted/dataset ratio for each patient and comparing the resulting distributions between real and synthetic populations (Fig. 6, bottom row). These ratio distributions capture how the ODE model processes each patient’s factor levels: if SA had generated patients with biologically implausible factor combinations, the model would process them differently, producing a shifted or broadened ratio distribution. Instead, the distributions overlapped substantially, with cloud overlap (fraction of synthetic ratios within the 5th–95th percentile of real ratios) ranging from 83% for ETP to 92% for lagtime, and two-sample Kolmogorov–Smirnov (KS) tests confirmed that the ratio distributions were statistically indistinguishable across all five TGA features ($D = 0.081\text{--}0.123$, all $p > 0.35$). For consistency with the bootstrap approach used at the previous level, we also subsampled the synthetic ratios to n_{real} and repeated the KS test 1,000 times: all five features achieved $p > 0.05$ in $\geq 96\%$ of replicates. Under TF+TM conditions, the model systematically overcorrected the protein C anticoagulant pathway, producing peak and maximum rate predictions approximately $0.5\times$ measured values for both real and synthetic patients (Supplementary Fig. S5), yet this shared systematic bias was itself informative: cloud overlap remained high at 88–92%, confirming that the model processed real and synthetic patients identically even under conditions where the calibration was imperfect.

The calibration also generalized across pregnancy timepoints: although fitted only on Visit 1 real patients, the predicted-to-measured ratios at Visits 2 and 3 under TF-only conditions were $0.96\times$ and $1.07\times$, respectively (Supplementary Fig. S6), confirming that the calibrated rate constants captured a stable population-level property of the coagulation system rather than overfitting to the training visit. Spearman rank correlations between predicted and measured values were moderate for ETP ($\rho \approx 0.6\text{--}0.8$ for real, $\rho \approx 0.6\text{--}0.65$ for synthetic; Supplementary Fig. S7) and weak for other features, consistent with a 5-parameter population-level calibration that was not designed to resolve inter-patient variability. The key finding was not patient-level prediction but population-level plausibility: an independent mechanistic model of the coagulation cascade confirmed that SA-generated patients fell within the same biologically plausible envelope as real patients, and could not be distinguished from them by the ODE model under either experimental condition.

2.5 Downstream Utility: Mechanistic Model Calibration

The preceding results established that SA-generated patients are statistically and mechanistically plausible. To test whether this plausibility translates to practical utility, we calibrated a mechanistic model entirely on synthetic patients and evaluated whether it could predict real patient outcomes. We calibrated the BZ2012 model (TF-only condition, 5 rate constants) on synthetic V1 patients ($N=100$) using the same bounded Nelder–Mead optimization applied to the real V1 calibration ($K=23$), with 11 random restarts to mitigate local-minimum sensitivity (Supplementary Table S4 for parameter details). We then evaluated both calibrations on held-out real V2 and V3 patients that neither model had seen during training. The synth-calibrated model achieved comparable or slightly better performance across all five TGA features on the held-out real patients, with per-feature median relative errors 2–10% lower than the real-calibrated model (overall ratio $0.94\times$; Fig. 7). This improvement likely reflects the smoother loss landscape afforded by $N=100$ synthetic training patients compared with $K=23$ real ones, reducing overfitting during optimization. The

two calibrations found different parameter values but produced highly correlated predictions across all five TGA features, confirming that the synthetic data captured the same underlying coagulation structure. We note that the synthetic training data is one step removed from the real data (real $\rightarrow$ SA $\rightarrow$ synthetic $\rightarrow$ calibration), so the synth-calibrated model is not fully independent of the real data; rather, it demonstrates that SA’s representation preserves sufficient biological structure for a mechanistic model to learn generalizable parameters.

3 Discussion

This study tested whether a geometry-preserving generative method could produce synthetic patients from a very small longitudinal cohort that were not merely statistically similar to real patients but consistent with the underlying coagulation biology. Our results provided evidence at each validation level that SA-generated patients met this standard. Individual feature distributions were reproduced with median MRE of 1.4%, which MVN also achieved by construction. SA additionally preserved the joint cross-visit covariance structure that MVN’s rank-deficient estimate could not represent, a difference made visible in the eigenvalue spectrum where MVN introduced spurious variance in 194 null dimensions while SA respected the low-rank geometry of the data. SA could amplify clinically meaningful rare subgroups, generating 100 synthetic PCOS patients from only 3 real ones, while preserving subgroup-specific signatures that MVN cannot capture. And an independent ODE model of the coagulation cascade, calibrated exclusively on real patients, confirmed that the factor-to-thrombin-generation mapping in synthetic patients was biologically plausible, with 83–92% cloud overlap and Kolmogorov–Smirnov tests unable to distinguish the two populations ($p > 0.35$ for all five TGA features).

The ability to generate validated synthetic cohorts from very small longitudinal datasets has practical implications for maternal health research. Rare pregnancy complications such as PCOS, preeclampsia, and gestational thrombophilia are difficult to study because assembling large, well-characterized longitudinal cohorts requires years of recruitment across multiple clinical sites. Our results suggest that SA can enable hypothesis generation, power analyses, and computational modeling studies for these understudied populations by generating synthetic cohorts that preserve both the statistical and mechanistic properties of the real data. The conditional generation capability is particularly relevant: rather than requiring researchers to collect additional rare patients, SA can amplify existing small subgroups into cohorts large enough for meaningful analysis.

Understanding why SA succeeds where parametric methods fail requires examining the geometric structure of the problem. SA generates samples by interpolating between stored patient profiles via the Hopfield energy landscape operating in a PCA-reduced linear subspace, rather than fitting a parametric distribution that must regularize or truncate high-dimensional structure. In the $n < p$ regime that characterizes most small clinical studies, any parametric distribution faces a fundamental identifiability problem: there are more parameters to estimate than observations to estimate them from. SA sidesteps this by operating in a PCA-reduced space where the memory-to-dimension ratio is favorable ($K/d_{\text{PCA}} \approx 1.28$) and by preserving the full directional structure of the data through the attention mechanism. A natural concern is whether the β^* selection procedure, originally developed for protein sequence generation [Varner, 2026b], transfers to clinical coagulation data. The entropy inflection method identifies the phase transition in the Hopfield energy landscape, which is a geometric property of K patterns in d dimensions, not a property of what the features represent; for our dataset the empirical inflection ($\beta^* \approx 2.94$) was close to the theoretical prediction ($\sqrt{d} \approx 4.24$), and a sensitivity analysis confirmed that generation quality degraded gracefully across $\beta/\beta^* \in [0.1, 3.0]$ (Supplementary Table S1). Similarly, varying the PCA

variance retention threshold from 85% to 99% ($d_{\text{PCA}} = 13$ to 21) had minimal impact on generation quality: median MRE remained between 1.2% and 1.8% across all thresholds (Supplementary Table S7), confirming that the 95% threshold was not a critical design choice. The multiplicity weighting for conditional generation introduced an additional consideration: achieving 80% attention on the PCOS subgroup required $\rho \approx 26.7$, reducing the effective pattern count to $K_{\text{eff}} \approx 4.6$ (Supplementary Table S2), which explains the larger MREs observed for PCOS features because the energy landscape was effectively shaped by fewer than 5 independent patterns, pulling means toward the population average. The direction-magnitude decomposition compounds this limitation: for PCOS, magnitudes are drawn from only 3 empirical norms, providing minimal diversity in the scale of generated samples. More broadly, the SA framework and its multiplicity-weighted conditioning have been independently validated on discrete protein sequence generation from small family alignments [Varner, 2026b], on steering generation toward functional subsets such as binding peptides [Varner, 2026a], and on the theoretical foundations connecting modern Hopfield networks to attention-based generation [Alswaidan and Varner, 2026]. Across these domains, the critical temperature prediction ($\beta^* \approx \sqrt{d}$) and the multiplicity-weighted conditioning mechanism transfer without modification, suggesting that the geometric properties exploited here are not specific to clinical coagulation data but are general features of the Hopfield energy landscape operating on small pattern sets.

Beyond these statistical and geometric considerations, the mechanistic validation introduced an approach that applies beyond coagulation: rather than relying solely on statistical comparisons between real and synthetic distributions, we tested whether synthetic patients produced biologically consistent outputs when processed by an independent computational model, and went further to show that a mechanistic model calibrated entirely on synthetic data predicted real patient outcomes as well as one calibrated on real data (overall ratio 0.94 $\times$ on held-out V2+V3 patients, with 11 random restarts per calibration). This approach is available whenever a validated mechanistic model exists for the domain of interest (pharmacokinetic models in drug development, physiological models in critical care, or metabolic models in oncology), and the requirement is not that the model be a perfect predictor but that synthetic patients fall within the same model-predicted envelope as real patients. The bootstrap Mann–Whitney analysis of condition-specific features revealed that 5 of 24 (21%) feature–condition pairs were statistically distinguishable at equal sample sizes, but the mean relative errors for these features remained small (5–15%), and the detected differences likely reflected SA’s tendency to compress distributional tails rather than shift means. This tail compression arises from two sources: the PCA dimensionality reduction, which cannot fully represent higher-order moments of every feature, and the direction-magnitude decomposition, which draws magnitudes from a finite empirical distribution of $K=23$ norms. This is a trade-off: SA sacrifices some tail fidelity in exchange for preserving the manifold geometry that parametric methods cannot capture in the $n < p$ regime.

To assess whether more expressive generative models could match SA at this sample size, we compared against CTGAN and TVAE [Xu et al., 2019], two widely used deep generative methods for tabular data (Supplementary Table S6). CTGAN failed completely, producing median MREs of $\approx 19\%$, an order of magnitude worse than SA, across all epoch counts tested (300–3,000), consistent with known GAN mode collapse behavior on small datasets. TVAE achieved comparable marginal fidelity to SA at high epoch counts (median MRE 1.8% at 3,000 epochs), but operated on per-visit rows ($n=90$) rather than concatenated longitudinal profiles ($n=23$), meaning it could not capture cross-visit covariance structure and had no mechanism for conditional subgroup generation. These results confirmed that SA’s advantage at small n is practical: standard deep generative methods either fail outright (CTGAN) or achieve marginal-level fidelity without the longitudinal and conditional capabilities that SA provides.

Several additional limitations warrant discussion. Our study used a single longitudinal dataset of $K=23$ patients, and further evaluation on independent datasets and across different clinical domains is needed to establish generalizability. The PCA dimensionality reduction assumed linear structure in the feature space; nonlinear manifold learning methods may better capture complex clinical relationships, though at the cost of reduced interpretability. The mechanistic validation under TF+TM conditions revealed systematic ODE model bias, though the observation that real and synthetic patients shared the same bias pattern was itself informative. The BZ2012 calibration required substantial departures from published rate constants, notably a 50-fold reduction in intrinsic tenase k_{cat} and a 15-fold increase in extrinsic tenase k_{cat} (Supplementary Table S4). These deviations likely reflect differences between the *in vitro* conditions under which the literature rate constants were measured (purified component systems) and the plasma-based TGA used in this study, where phospholipid surfaces, factor concentrations, and inhibitor profiles differ from the model assumptions [Hockin et al., 2002, Brummel-Ziedins et al., 2012]. The purpose of the mechanistic validation was not to obtain biochemically accurate rate constants but to test whether the ODE model processed real and synthetic patients equivalently, a comparison that is invariant to the absolute parameter values. The mechanistic validation was limited to thrombin generation (BZ2012) and did not cover the fibrinolytic or viscoelastic features included in the dataset; extending this validation to a mechanistic model of fibrinolysis, currently under development, would provide additional evidence for these feature categories. Finally, we did not validate synthetic patients against clinical outcomes; our validation was restricted to statistical and mechanistic plausibility, and prospective validation against real clinical decision-making remains an important future direction.

4 Conclusion

We have demonstrated that multiplicity-weighted Stochastic Attention can generate synthetic longitudinal patient data from very small clinical cohorts ($K=23$) that pass four levels of validation: marginal plausibility, cross-visit covariance preservation, conditional subgroup amplification, and mechanistic consistency with an independent ODE model of the coagulation cascade. A downstream utility test showed that a mechanistic model calibrated entirely on synthetic patients predicted real patient outcomes as well as one calibrated on real data. The SA framework preserves the geometry of the clinical data in regimes where parametric approaches such as multivariate normal sampling fail due to rank deficiency. Combined with a multi-level validation framework, this approach provides a practical path for generating clinically useful synthetic cohorts to support computational research in rare conditions and small longitudinal studies.

5 Methods

5.1 Population Dataset

We analyzed a longitudinal dataset of coagulation, fibrinolysis, thrombin generation assay (TGA), and rotational thromboelastometry (ROTEM) as secondary analysis measurements from samples collected longitudinally from women desiring a pregnancy in a study that was approved by the Institutional Review Board at the University of Vermont. Written consent was obtained from all participants prior to enrollment. Blood collection, plasma processing, and assay protocols have been described previously [McLean et al., 2012, Hale et al., 2012, Psinos et al., 2021, Bernstein et al., 2016]. Briefly, citrated platelet-poor plasma was collected without tourniquet use after supine

rest at three clinical visits: Visit 1 (pre-pregnancy baseline in the follicular phase), Visit 2 (end of first trimester), and Visit 3 (mid-third trimester). Plasma aliquots were stored at -80°C for subsequent analysis. Measurements of secondary analytes were conducted as follows: activity levels of coagulation factors were determined by Stago clotting assays, fibrinolysis proteome levels were measured by ELISA, and hormone levels were determined by the CLIA-certified laboratory at the University of Vermont Medical Center. TGA was performed using 5 pM relipidated tissue factor with fluorogenic substrate on a Synergy4 plate reader as described in McLean et al. [2012]. Clot formation and fibrinolysis dynamics were assessed via viscoelastometry using the ROTEM Delta Instrument (Werfen). Thawed plasma was mixed with or without 4 nM tPA (with respect to plasma volume), recalcified with 15 mM calcium chloride, and incubated for 3 minutes at 37°C . The reaction was initiated by adding the plasma mixture to ROTEM cups containing tissue factor at a 17:1 ratio; the final concentration of TF in the reaction was 5 pM.

The dataset contained 153 total records across approximately 50 subjects. After removing columns with $> 30\%$ missing values and retaining only subjects with complete data across all three visits, we obtained $K = 23$ complete longitudinal profiles across $n = 72$ assay measurements per visit (Table 1). The 72 assay measurements spanned five categories: (i) hormones (estradiol, progesterone), (ii) coagulation factors and inhibitors (Factors II, V, VII, VIII, IX, X, XI, XII; AT, PC, TFPI, vWF, fibrinogen, etc.), (iii) fibrinolytic markers (plasminogen, PAI-1, PAI-2, α_2 -AP, TAFI, tPA, D-dimer equivalents), (iv) thrombin generation parameters under four initiator conditions (TF at 5 pM, TF+TM, No TF, No TF+anti-FXIa), and (v) ROTEM/viscoelastic parameters (CT, MCF, alpha angle, LOT, ML, LT, AUC under TF Only and TF+tPA conditions).

The dataset included three enrollment conditions: Healthy nulliparous women ($n=14$), women with a personal history of preeclampsia from a previous pregnancy (Prior PE, $n=6$), and women with polycystic ovarian syndrome (PCOS, $n=3$; 2 individuals were nulliparous). For conditional generation, we defined three overlapping subgroups based on pregnancy outcome and comorbidity rather than enrollment condition: Uncomplicated ($n=18$, all patients whose study pregnancy did not result in preeclampsia), PCOS ($n=3$, a cross-cutting comorbidity; 1 patient also developed PE), and Developed PE ($n=5$, patients who developed preeclampsia during the study pregnancy, drawn from 2 healthy-enrolled, 2 prior-PE-enrolled, and 1 PCOS-enrolled participants).

5.2 Multiplicity-Weighted Stochastic Attention

We generated synthetic patients using a multiplicity-weighted variant of Stochastic Attention (SA) rooted in modern Hopfield network theory [Ramsauer et al., 2021]. We stored K memory patterns as columns of a matrix $\hat{\mathbf{X}} \in \mathbb{R}^{d \times K}$ and defined a weighted Hopfield energy landscape:

$$E_r(\boldsymbol{\xi}) = \frac{1}{2} \|\boldsymbol{\xi}\|^2 - \frac{1}{\beta} \log \sum_{k=1}^K r_k \exp\left(\beta \mathbf{m}_k^\top \boldsymbol{\xi}\right) \quad (1)$$

where $\{\mathbf{m}_k\}_{k=1}^K$ are stored memory patterns, β is an inverse temperature parameter controlling the sharpness of retrieval, and $\{r_k\}_{k=1}^K$ are per-pattern multiplicity weights. When all $r_k = 1$, this reduces to the standard modern Hopfield energy; when $r_k = \rho > 1$ for a designated subset of patterns, the energy landscape is continuously deformed to favor that subset. We sampled from this landscape using an Unadjusted Langevin Algorithm (ULA):

$$\boldsymbol{\xi}_{t+1} = (1 - \alpha) \boldsymbol{\xi}_t + \alpha \hat{\mathbf{X}} \text{softmax}\left(\beta \hat{\mathbf{X}}^\top \boldsymbol{\xi}_t + \log \mathbf{r}\right) + \sqrt{\frac{2\alpha}{\beta}} \boldsymbol{\varepsilon}_t \quad (2)$$

where α is a step size, $\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and $\log \mathbf{r}$ is the element-wise log of the multiplicity vector. The deterministic component drove $\boldsymbol{\xi}_t$ toward a multiplicity-weighted combination of stored patterns,

while the stochastic component enabled exploration and novelty. The $\log r_k$ bias in the softmax logits provided a minimal, theoretically grounded mechanism for continuously interpolating between unconditioned generation ($\rho = 1$, all patterns equally weighted) and hard subset curation ($\rho \rightarrow \infty$, only designated patterns contribute). This is an inference-time operation that requires no retraining of the energy landscape.

The inverse temperature β controlled the trade-off between retrieval (large β , converging to nearest memory) and generation (small β , uniform mixing). We identified the critical β^* at the phase transition by computing the normalized attention entropy and finding the inflection point (maximum negative second derivative of $\bar{H}(\beta)$ in $\log\text{-}\beta$ space) over a logarithmic sweep of 80 values from $\beta = 0.1$ to 1000. This procedure is domain-agnostic: the phase transition is a geometric property of K patterns in d dimensions, not a property of the features they represent. For our dataset ($K=23$, $d_{\text{PCA}}=18$), the empirical inflection yielded $\beta^* \approx 2.94$, close to the theoretical prediction $\beta^* = \sqrt{d} \approx 4.24$. A sensitivity analysis across $\beta/\beta^* \in [0.1, 3.0]$ confirmed that the operating point balanced generation novelty against fidelity (Supplementary Table S1). For weighted sampling, the entropy calculation incorporated the multiplicity bias, yielding a ρ -dependent $\beta^*(\rho)$.

The effective number of patterns contributing to the dynamics was quantified by the participation ratio $K_{\text{eff}} = (\sum_k r_k)^2 / \sum_k r_k^2$, which interpolated smoothly between the total pattern count (unconditioned) and the designated subset size (strongly conditioned). For a target effective fraction f_{target} of probability mass on the designated subset, the required multiplicity was $\rho = f_{\text{target}} \cdot K_{\text{bg}} / (K_{\text{des}} \cdot (1 - f_{\text{target}}))$, where K_{des} and K_{bg} are the designated and background pattern counts, respectively.

5.3 Pipeline for Continuous Longitudinal Data

Applying SA to continuous longitudinal clinical data required several adaptations beyond the standard Hopfield sampling framework. We first concatenated each patient’s $n=72$ assay measurements across all three visits into a single vector of dimension $d_{\text{concat}} = 3n = 216$, so that each stored memory pattern encoded a complete longitudinal profile and generated synthetic patients would have internally consistent multi-visit trajectories. We then standardized each feature to zero mean and unit variance and applied PCA retaining 95% of the variance, reducing dimensionality from 216 to $d_{\text{PCA}}=18$. This compression served two purposes: it removed collinearity among the 216 features and yielded a memory-to-dimension ratio of $K/d_{\text{PCA}} \approx 1.28$, placing SA in a favorable operating regime where the number of stored patterns exceeded the dimensionality of the representation.

Standard SA operates on unit-norm patterns, which is appropriate for discrete data (e.g., one-hot protein sequences) but destroys the anisotropic variance structure of continuous clinical measurements, collapsing dispersion across all principal components to a unit sphere. We addressed this with a direction-magnitude decomposition: before normalization, we recorded each pattern’s PCA-space norm $r_k = \|\mathbf{m}_k\|$; we then ran SA on unit-norm patterns to obtain a directional sample $\hat{\xi}$, drew a magnitude r from the empirical distribution of $\{r_k\}$, and reconstructed the rescaled sample as $\xi_{\text{rescaled}} = r \cdot \hat{\xi}$. This preserved the directional structure learned by the Hopfield energy while restoring the natural scale of variation. The rescaled PCA vector was mapped back to the original feature space via inverse PCA and destandardization, then split into three 72-dimensional visit records.

To generate condition-specific cohorts (e.g., PCOS-only), we set the multiplicity $r_k = \rho$ for patterns belonging to the target subgroup and $r_k = 1$ for all others, and sampled using the weighted Langevin update (Eq. 2). The multiplicity ρ was computed to achieve a target effective fraction $f_{\text{target}} = 0.80$ of softmax attention on the designated subset, chosen to strongly bias generation toward the target subgroup while retaining 20% background influence for interpolation diversity,

via $\rho = f_{\text{target}} \cdot K_{\text{bg}} / (K_{\text{des}} \cdot (1 - f_{\text{target}}))$. For the PCOS subgroup ($K_{\text{des}}=3$), this required $\rho \approx 26.7$, reducing the effective pattern count to $K_{\text{eff}} \approx 4.6$ (Supplementary Table S2). Magnitudes for the direction-magnitude decomposition were drawn from the condition-specific subset of norms, ensuring that the scale of variation matched the target subpopulation rather than the full cohort.

The complete generation pipeline is summarized in Algorithm 1. All experiments used step size $\alpha = 0.01$, $T = 2,000$ Langevin iterations per sample, and random seed 42 for reproducibility. At this step size, the Unadjusted Langevin Algorithm is valid without a Metropolis–Hastings acceptance correction. The pipeline was implemented in Julia (v1.12) using the packages listed in the code repository. Generating 100 synthetic patients required approximately 0.1 seconds on a standard laptop (Apple M-series), comparable to MVN sampling (≈ 0.15 s), because SA operates in the 18-dimensional PCA space rather than the full 216-dimensional feature space. Each synthetic patient was generated by an independent Langevin chain (no shared state between patients); generation quality was stable across $T \in [500, 4000]$ iterations (median MRE 1.0–1.6%), confirming that $T=2,000$ was sufficient for convergence.

Algorithm 1 Multiplicity-Weighted SA for Longitudinal Patient Generation

Require: Patient data $\{\mathbf{x}_k^{(v)}\}$ for $k = 1, \dots, K$ patients, $v = 1, 2, 3$ visits

Require: Multiplicity vector $\mathbf{r}$ (all ones for unconditioned; $r_k = \rho$ for designated subset)

Ensure: N synthetic longitudinal patient profiles

- 1: Concatenate: $\mathbf{x}_k \leftarrow [\mathbf{x}_k^{(1)}; \mathbf{x}_k^{(2)}; \mathbf{x}_k^{(3)}] \in \mathbb{R}^{216}$
 - 2: Standardize: $\mathbf{z}_k \leftarrow (\mathbf{x}_k - \boldsymbol{\mu}) \oslash \boldsymbol{\sigma}$
 - 3: PCA: $\mathbf{m}_k \leftarrow \mathbf{W}^\top \mathbf{z}_k \in \mathbb{R}^{18}$ ▷ 95% variance retained
 - 4: Record norms: $r_k^{\text{mag}} \leftarrow \|\mathbf{m}_k\|$; normalize $\hat{\mathbf{m}}_k \leftarrow \mathbf{m}_k / r_k^{\text{mag}}$
 - 5: Find β^* via entropy inflection on $\hat{\mathbf{M}} = [\hat{\mathbf{m}}_1, \dots, \hat{\mathbf{m}}_K]$
 - 6: **for** $i = 1, \dots, N$ **do**
 - 7: Initialize $\hat{\boldsymbol{\xi}}_0 \sim \text{Uniform}(\mathcal{S}^{17})$ ▷ random point on unit sphere
 - 8: **for** $t = 0, \dots, T - 1$ **do** ▷ Langevin dynamics
 - 9: $\hat{\boldsymbol{\xi}}_{t+1} \leftarrow (1 - \alpha)\hat{\boldsymbol{\xi}}_t + \alpha \hat{\mathbf{M}} \text{softmax}(\beta^* \hat{\mathbf{M}}^\top \hat{\boldsymbol{\xi}}_t + \log \mathbf{r}) + \sqrt{2\alpha/\beta^*} \boldsymbol{\varepsilon}_t$
 - 10: **end for**
 - 11: Draw magnitude: $r \sim \{r_k^{\text{mag}}\}$ ▷ condition-specific if weighted
 - 12: Rescale: $\mathbf{m}_{\text{synth}} \leftarrow r \cdot \hat{\boldsymbol{\xi}}_T / \|\hat{\boldsymbol{\xi}}_T\|$
 - 13: Inverse PCA + destandardize: $\mathbf{x}_{\text{synth}} \leftarrow \boldsymbol{\sigma} \odot (\mathbf{W} \mathbf{m}_{\text{synth}} + \bar{\mathbf{z}}) + \boldsymbol{\mu}$
 - 14: Split into visit records: $\mathbf{x}_{\text{synth}}^{(1)}, \mathbf{x}_{\text{synth}}^{(2)}, \mathbf{x}_{\text{synth}}^{(3)}$
 - 15: **end for**
-

5.4 Baseline and Validation Framework

As a baseline, we generated synthetic patients by fitting a single multivariate normal (MVN) distribution to the same $K=23$ concatenated 216-dimensional patient profiles. Critically, both SA and MVN operated on the same concatenated representation: each patient’s three visits were joined into one vector before generation, so that both methods had the opportunity to capture cross-visit dependencies. For SA, this structure was preserved through the Hopfield energy landscape operating on 18-dimensional PCA projections of the concatenated vectors. For MVN, the 216×216 sample covariance matrix had rank at most 22 and was regularized using Ledoit–Wolf shrinkage [Ledoit and Wolf, 2004], which inflated eigenvalues in the 194 null dimensions. We then evaluated both SA and MVN synthetic patients through four progressively stringent validation levels. **Level 1 (Marginal**

Plausibility) tested whether individual feature distributions matched, using per-feature means, standard deviations, and known physiological relationships. **Level 2 (Joint Structure)** tested whether the cross-visit covariance was preserved, by comparing full 216×216 correlation matrices, eigenvalue spectra, and PCA projections between real, SA, and MVN populations. **Level 3 (Conditional Structure)** tested whether rare subgroups could be amplified while preserving condition-specific signatures, by generating cohorts of 100 patients each from the Uncomplicated ($n=18$), PCOS ($n=3$), and Developed PE ($n=5$) subgroups using multiplicity-weighted sampling. **Level 4 (Mechanistic Consistency)** tested whether synthetic patients produced biologically plausible outputs under an independent mechanistic model. We used the Hockin–Mann BZ2012 coagulation model [Hockin et al., 2002, Brummel-Ziedins et al., 2012], a system of 58 ODEs with 64 rate constants, in which 5 rate constants were calibrated on Visit 1 real patients at the population level and the remaining 59 were held at literature values. We ran the model on all real and synthetic patients under two conditions (TF-only and TF+TM) and compared predicted-versus-measured thrombin generation features, using cloud overlap, defined as the fraction of synthetic predicted/measured ratios falling within the 5th–95th percentile range of real ratios, as the primary metric.

Data Availability

The code for synthetic patient generation, all validation scripts, and the generated synthetic datasets are available at <https://github.com/varnerlab/SA-generation-legacy-dataset-paper>. The real patient data were collected under approval from the University of Vermont Institutional Review Board; de-identified data are available upon reasonable request to the corresponding author.

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Author Contributions

J.D.V. conceived the study, developed the SA pipeline and validation framework, performed computational analyses, and wrote the manuscript. M.C.B. and T.O. oversaw the secondary analysis measurements of coagulation and fibrinolysis markers, dynamic assays, and hormone measurements; provided domain expertise on coagulation biology and thrombin generation assays; and reviewed the manuscript. C.M. was involved in the enrollment of participants in the prospective research study and collection of clinical information of the participants. I.B. designed and oversaw the primary prospective research study of enrolled participants, provided clinical interpretation, and reviewed the manuscript. All authors approved the final version.

Competing Interests

The authors declare no competing interests.

References

- Sarah Toy. U.S. maternal mortality hits highest level since 1965. *The Wall Street Journal*, 2023. March 16, 2023.
- Emily Harris. US maternal mortality continues to worsen. *JAMA*, 329(15):1248, 2023. doi: 10.1001/jama.2023.5254.
- Susanna L Trost, Jennifer Beauregard, Fanta Njie, Joyita Berry, Allison Harvey, David A Goodman, Emily E Petersen, and Shanna Cox. Pregnancy-related deaths: Data from maternal mortality review committees in 36 US states, 2017–2019, 2022. <https://www.cdc.gov/maternal-mortality/php/data-research/>.
- Kenneth G Mann, Kathleen Brummel-Ziedins, Thomas Orfeo, and Saulius Butenas. Thrombin generation in hemorrhage control and vascular occlusion. *Circulation*, 124(2):225–235, 2011. doi: 10.1161/CIRCULATIONAHA.110.952648.
- Saulius Butenas, Thomas Orfeo, and Kenneth G Mann. Tissue factor in coagulation: Which? where? when? *Arteriosclerosis, Thrombosis, and Vascular Biology*, 29(12):1989–1996, 2009. doi: 10.1161/ATVBAHA.108.177402.
- Mark F Hockin, Kenneth C Jones, Stephen J Everse, and Kenneth G Mann. A model for the stoichiometric regulation of blood coagulation. *Journal of Biological Chemistry*, 277(21):18322–18333, 2002. doi: 10.1074/jbc.M201173200.
- Margareta Hellgren. Hemostasis during normal pregnancy and puerperium. *Seminars in Thrombosis and Hemostasis*, 29(2):125–130, 2003. doi: 10.1055/s-2003-38897.
- E Grouzi, A Pouliakis, A Aktypi, et al. Pregnancy and thrombosis risk for women without a history of thrombotic events: a retrospective study of the real risks. *Thrombosis Journal*, 20(60), 2022. doi: 10.1186/s12959-022-00419-6.
- Katarina A Bremme. Haemostatic changes in pregnancy. *Best Practice & Research Clinical Haematology*, 16(2):153–168, 2003. doi: 10.1016/S1521-6926(03)00021-5.
- Ricardo Azziz, Keslie S Woods, Rosario Reyna, Thomas J Key, Eric S Knochenhauer, and Bulent O Yildiz. The prevalence and features of the polycystic ovary syndrome in an unselected population. *Journal of Clinical Endocrinology & Metabolism*, 89(6):2745–2749, 2004. doi: 10.1210/jc.2003-032046.
- Edgardo Abalos, Cristina Cuesta, Guillermo Carroli, Zahida Qureshi, Mariana Widmer, Joshua P Vogel, and João Paulo Souza. Pre-eclampsia, eclampsia and adverse maternal and perinatal outcomes: a secondary analysis of the World Health Organization multicountry survey on maternal and newborn health. *BJOG: An International Journal of Obstetrics & Gynaecology*, 121:14–24, 2014. doi: 10.1111/1471-0528.12629.
- Maria C Bravo, Thomas Orfeo, Kenneth G Mann, and Stephen J Everse. Modeling of human factor Va inactivation by activated protein C. *BMC Systems Biology*, 6:45, 2012. doi: 10.1186/1752-0509-6-45.

- Kathleen E Brummel-Ziedins, Thomas Orfeo, Peter W Callas, Matthew Gissel, Kenneth G Mann, and Edwin G Bovill. The prothrombotic phenotypes in familial protein c deficiency are differentiated by computational modeling of thrombin generation. *PLoS ONE*, 7(9):e44378, 2012. doi: 10.1371/journal.pone.0044378.
- Deyan Luan, Michael Zai, and Jeffrey D Varner. Computationally derived points of fragility of a human cascade are consistent with current therapeutic strategies. *PLoS Computational Biology*, 3(7):e142, 2007. doi: 10.1371/journal.pcbi.0030142.
- Deyan Luan, Fania Szlam, Kenichi A Tanaka, Philip S Barie, and Jeffrey D Varner. Ensembles of uncertain mathematical models can identify network response to therapeutic interventions. *Molecular BioSystems*, 6(11):2272–2286, 2010. doi: 10.1039/b920693k.
- Adarsh Gonzales, Guruprabha Guruswamy, and Sandesh R Smith. Synthetic data generation methods in healthcare: A review on open-source tools and methods. *PLOS Digital Health*, 2(1):e0000082, 2023. doi: 10.1371/journal.pdig.0000082.
- Hasan Murtaza, Nicholas Schreck, Jil Gao, Andreas D Meid, Thomas Gredner, Walter E Haefeli, Hermann Brenner, and Ben Schottker. Synthetic data generation for a longitudinal cohort study – evaluation, method extension and reproduction of published data analysis results. *Scientific Reports*, 14:12785, 2024. doi: 10.1038/s41598-024-62102-2.
- Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004. doi: 10.1016/S0047-259X(03)00096-4.
- Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. In *Advances in Neural Information Processing Systems*, volume 27, pages 2672–2680, 2014.
- Diederik P Kingma and Max Welling. Auto-encoding variational Bayes. In *International Conference on Learning Representations (ICLR)*, 2014.
- Lei Xu, Maria Skoularidou, Alfredo Cuesta-Infante, and Kalyan Veeramachaneni. Modeling tabular data using conditional GAN. 32, 2019.
- Hubert Ramsauer, Bernhard Schäfl, Johannes Lehner, Philipp Seidl, Michael Widrich, Thomas Adler, Lukas Gruber, Markus Holzleitner, Milena Pavlović, Geir Kjetil Sandve, Victor Greiff, David Kreil, Michael Kopp, Gunter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield networks is all you need. In *International Conference on Learning Representations (ICLR)*, 2021.
- Jeffrey D Varner. Conditioning protein generation via Hopfield pattern multiplicity. *arXiv preprint arXiv:2603.20115*, 2026a. doi: 10.48550/arXiv.2603.20115.
- Maria Bravo, Thomas Orfeo, Ira Bernstein, Sandra Vadhin, Divya Lakshmi, and Jeffrey D Varner. BSTModelKit.jl: Building biochemical systems theory models of individualized coagulation dynamics during pregnancy, 2023. JuliaCon 2023.
- Jeffrey D Varner. Training-free generation of protein sequences from small family alignments via stochastic attention. *arXiv preprint arXiv:2603.14717*, 2026b. doi: 10.48550/arXiv.2603.14717.

- 614 Abdulrahman Alswaidan and Jeffrey D Varner. Stochastic attention via Langevin dynamics on
615 the modern Hopfield energy. *arXiv preprint arXiv:2603.06875*, 2026. doi: 10.48550/arXiv.2603.
616 06875.
- 617 Kelley C McLean, Ira M Bernstein, and Kathleen Brummel-Ziedins. Tissue factor dependent
618 thrombin generation across pregnancy. *American Journal of Obstetrics and Gynecology*, 207(2):
619 135.e1–135.e6, 2012. doi: 10.1016/j.ajog.2012.05.027.
- 620 Sarah A Hale, Burton Sobel, Anna Benvenuto, Adrienne Schonberg, Gary J Badger, and Ira M
621 Bernstein. Coagulation and fibrinolytic system protein profiles in women with normal pregnancies
622 and pregnancies complicated by hypertension. *Pregnancy Hypertension*, 2(2):152–157, 2012. doi:
623 10.1016/j.preghy.2012.01.004.
- 624 Rachel B C Psinos, Erin A Morris, Carole A McBride, and Ira M Bernstein. Association of pre-
625 pregnancy subclinical insulin resistance with cardiac dysfunction in healthy nulliparous women.
626 *Pregnancy Hypertension*, 26:11–16, 2021. doi: 10.1016/j.preghy.2021.07.246.
- 627 Ira M Bernstein, Sarah A Hale, and Gary J Badger. Differences in cardiovascular function
628 comparing prior preeclampsia with nulliparous controls. *Pregnancy Hypertension*, 2016. doi:
629 10.1016/j.preghy.2016.11.001.

Table 1: **Demographics and clinical characteristics of the 23 patients with complete longitudinal data across all three visits.** Values are reported as mean $\pm$ SD, median (range), or n (%) as appropriate.

Characteristic	Mean $\pm$ SD	Median (Range)	n (%)
Age at enrollment (yrs)	30.2 $\pm$ 5.1	29 (20–41)	
Race			
White			22 (96)
Not disclosed			1 (4)
Prepregnancy BMI	26.6 $\pm$ 5.3	25 (19.0–37.8)	
Parity		0 (0–2)	
Nulliparous			16 (70)
Parous			7 (30)
Enrollment condition			
Healthy nulliparous			14 (61)
PCOS			3 (13)
Prior PE			6 (26)
Cycle day at V1	9.2 $\pm$ 3.6	10 (3–14)	
Gestational age at V2 (days)	89.0 $\pm$ 5.5		
Gestational age at V3 (days)	217.0 $\pm$ 4.0		
<i>Enrollment cohort $\times$ pregnancy outcome</i>			
	Uncomplicated	Developed PE	Total
Healthy nulliparous	12	2	14
Prior PE	4	2	6
PCOS	2	1	3
Total	18	5	23

Table 2: Per-visit mean relative error (MRE) between real and SA-generated synthetic patients for representative coagulation features, computed as $|\bar{x}_{\text{synth}} - \bar{x}_{\text{real}}|/|\bar{x}_{\text{real}}|$.

Feature	Visit 1	Visit 2	Visit 3
Factor II (%)	0.018	0.017	0.007
Factor VIII (%)	0.018	0.008	0.005
AT (%)	0.025	0.001	0.008
Fibrinogen (mg/dL)	0.005	0.009	<0.001
TF Peak (nM)	0.025	0.002	0.017
TF ETP (nM·min)	0.013	0.008	0.023

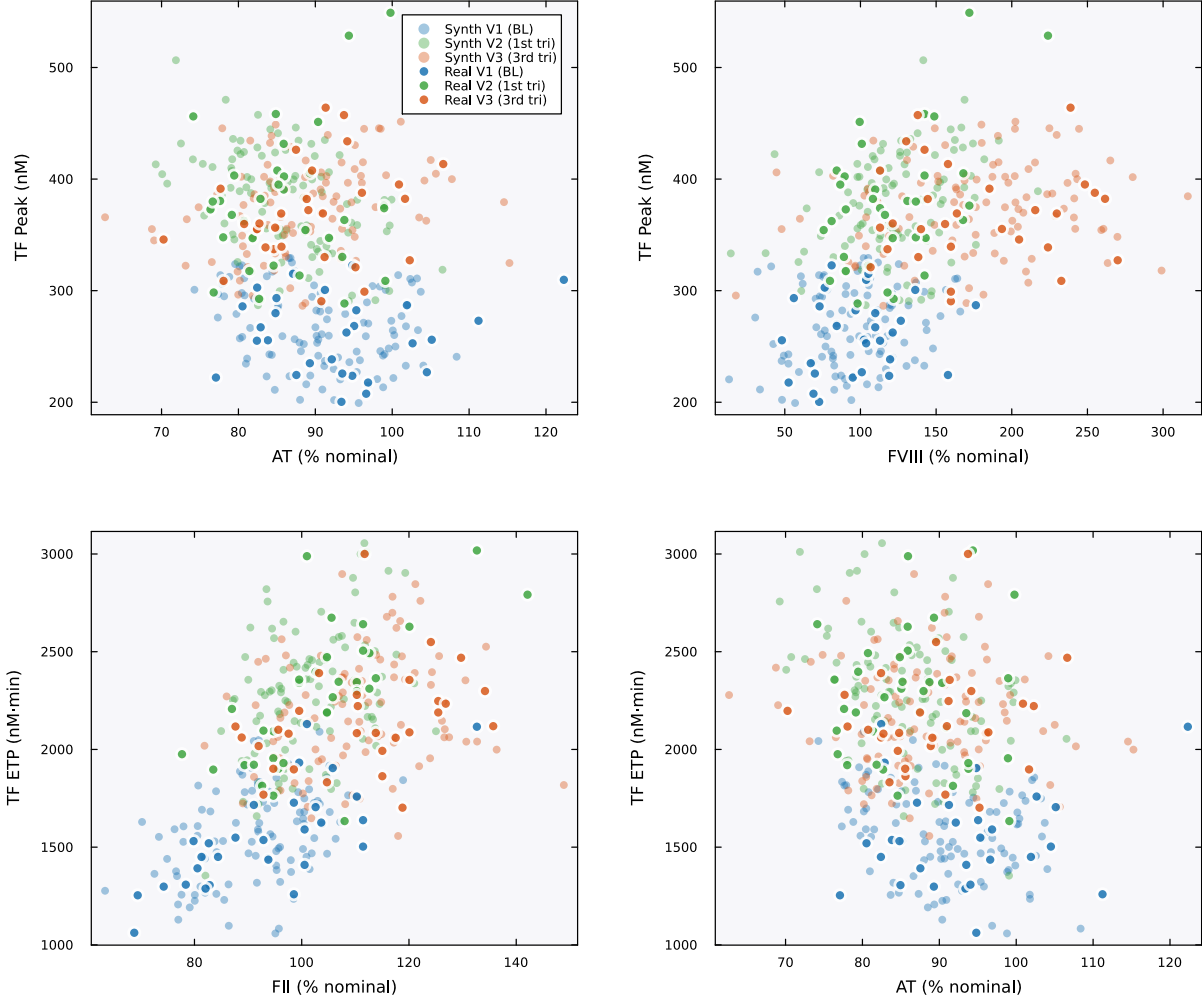


Figure 1: **Physiological correlations are preserved in synthetic patients.** Scatter plots of coagulation factor levels versus thrombin generation parameters across all three visits. Filled markers are real patients; open markers are SA-generated synthetic patients. Colors encode visit: blue (V1/baseline), green (V2/first trimester), orange (V3/third trimester). Real and synthetic patients occupy the same joint regions across all four pairwise relationships, confirming that biologically meaningful dependencies are preserved: AT inversely correlates with TF-initiated peak thrombin (top left), FVIII positively correlates with peak (top right), and both FII (bottom left) and AT (bottom right) show the expected relationships with ETP. The visit-dependent shift in these relationships, reflecting the pregnancy-driven increase in procoagulant factors, is also preserved in the synthetic cohort.

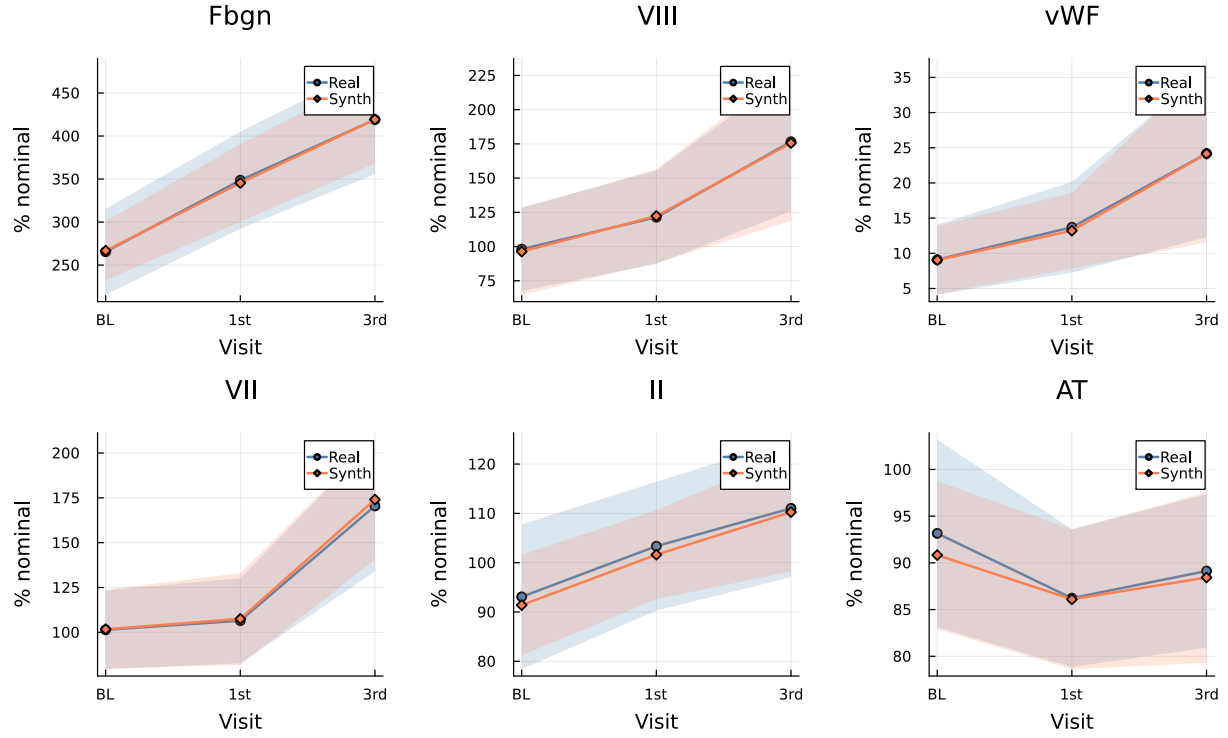


Figure 2: **Pregnancy-driven longitudinal trajectories are reproduced.** Mean $\pm$ standard deviation bands for six key coagulation features across visits (BL = baseline, 1st = first trimester, 3rd = third trimester) for real (blue) and SA-generated synthetic (orange) patients. The characteristic pregnancy-driven increases in fibrinogen, Factor VIII, and vWF are captured, as are the stable-to-declining patterns in Factor II and AT.

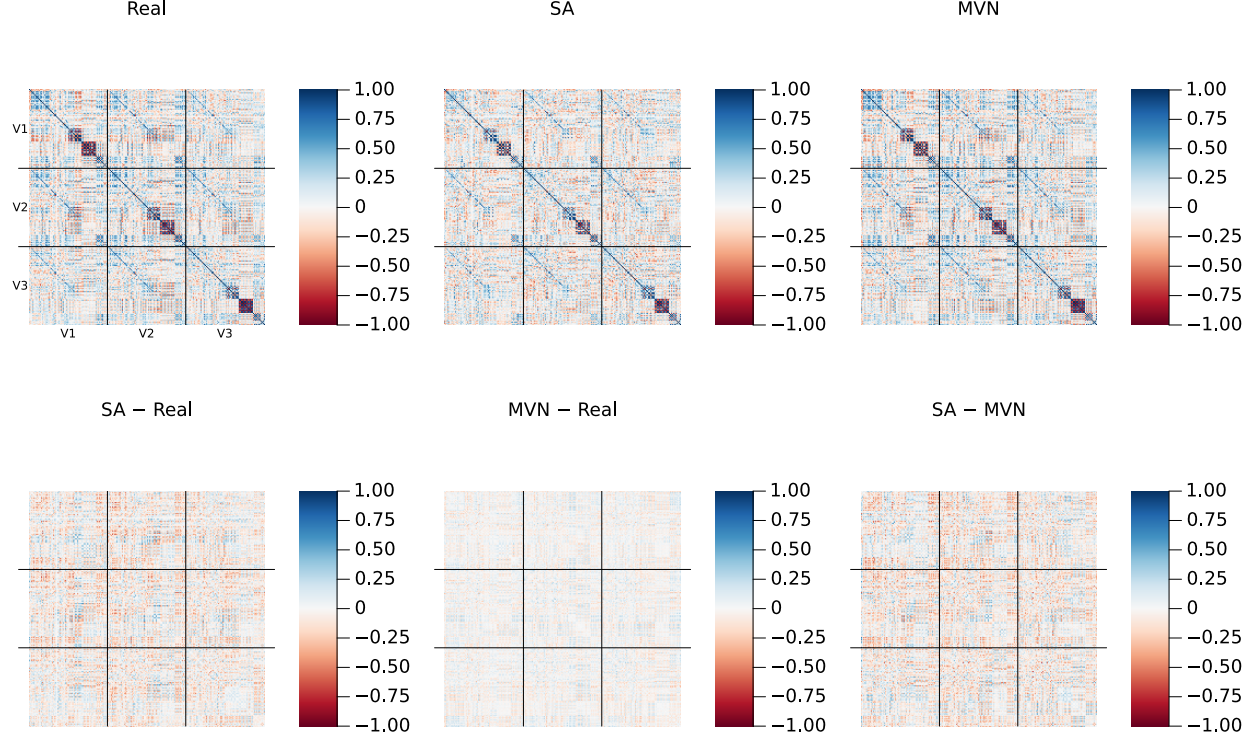


Figure 3: **Cross-visit correlation structure.** Top row: full 216×216 Pearson correlation matrices for Real ($K=23$, left), SA ($N=100$, center), and MVN ($N=100$, right) populations. Each matrix is organized as a 3×3 grid of 72×72 blocks (delineated by black lines), where the diagonal blocks capture within-visit feature correlations (V1–V1, V2–V2, V3–V3) and the off-diagonal blocks capture cross-visit dependencies (e.g., V1–V3: does a patient’s baseline Factor VIII predict their third-trimester Factor VIII?). SA preserves both the within-visit and cross-visit block structure. Bottom row: pairwise residual matrices (element-wise subtraction), all plotted on the same ± 1 color scale as the top row. The SA–Real residual shows small, unstructured deviations. The MVN–Real residual is smoother, particularly in the off-diagonal blocks, reflecting the Ledoit–Wolf regularization that systematically shrinks cross-visit correlations toward zero. The SA–MVN residual confirms that the two methods differ most in these off-diagonal blocks, where SA retains longitudinal dependencies that MVN’s rank-deficient covariance estimate cannot represent. A version with amplified residual color scale (± 0.5) and Frobenius norms is provided in Supplementary Fig. S1.

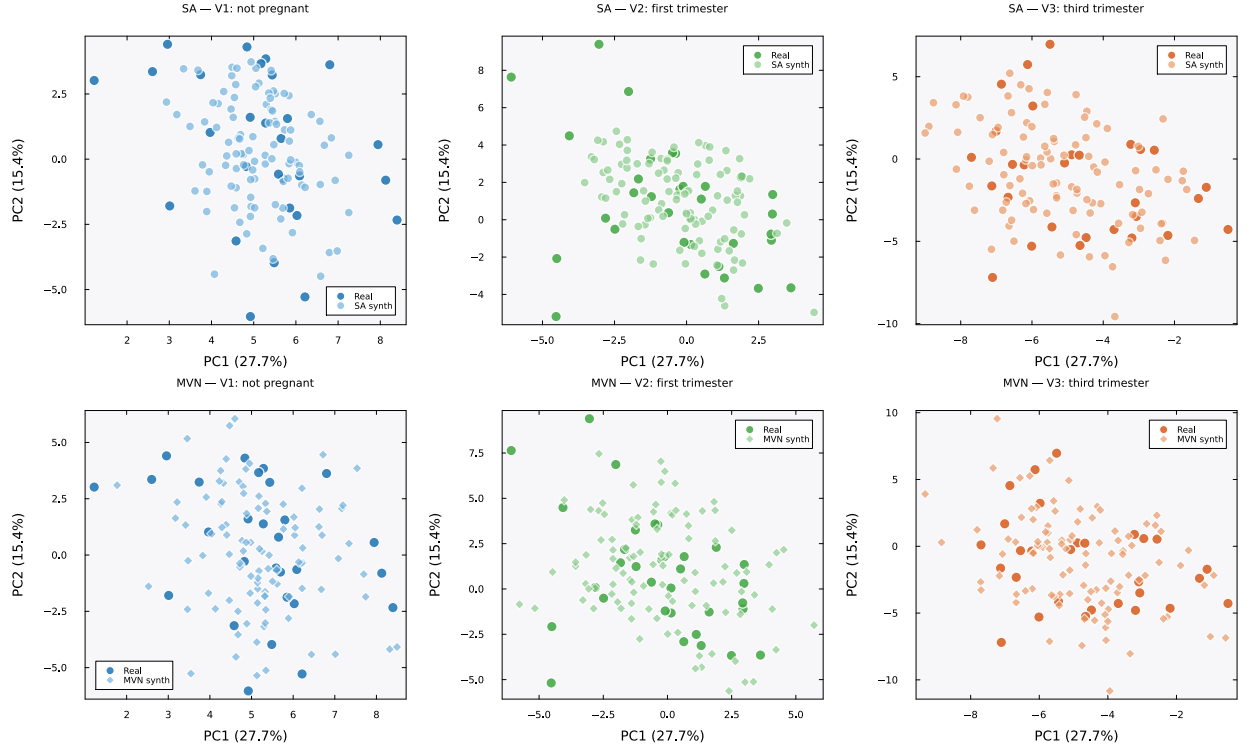


Figure 4: **PCA projections by visit: SA vs. MVN.** Each panel shows the first two principal components (PC1: 27.7% variance, PC2: 15.4%) computed from standardized per-visit real patient data ($K=23$ patients per visit, 72 features). Dark markers indicate real patients; lighter markers indicate synthetic patients ($N=100$). Top row: SA-generated patients (circles) cluster tightly around the real data cloud at all three visits. Bottom row: MVN-generated patients (diamonds) show greater dispersion, particularly at Visits 2 (first trimester) and 3 (third trimester), where MVN patients extend beyond the convex hull of the real data. This excess scatter reflects the spurious variance introduced by Ledoit–Wolf regularization of the rank-deficient covariance (Supplementary Fig. S2). Colors encode visit: blue (V1, not pregnant), green (V2, first trimester), orange (V3, third trimester).

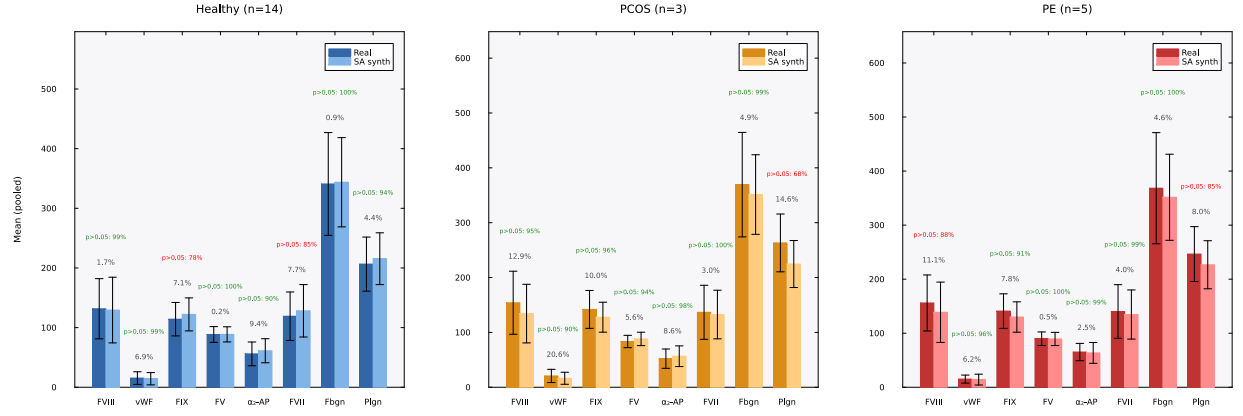


Figure 5: Condition-specific feature preservation. Grouped bar charts comparing real (dark bars) and SA-generated synthetic (light bars) patient means ($\pm$ SD error bars) for eight coagulation features, shown separately for each clinical subgroup: Uncomplicated ($n=18$, left, blue), PCOS ($n=3$, center, orange), and Developed PE ($n=5$, right, red). Values are pooled across all three visits. Gray percentages indicate the mean relative error (MRE). Green annotations show the bootstrap Mann–Whitney equivalence result: the fraction of 1,000 bootstrap replicates (synthetic subsampled to n_{real}) in which $p > 0.05$, i.e., real and synthetic could not be distinguished. Features achieving $\geq 90\%$ are shown in green; $< 90\%$ in red. Across all 24 feature–condition pairs, 19 (79%) were statistically indistinguishable in $\geq 90\%$ of replicates (median: 98.4%). The five features below this threshold were distributed across high-variance features (FIX, FVII, FVIII, plasminogen) where large inter-patient variability reduced test power.

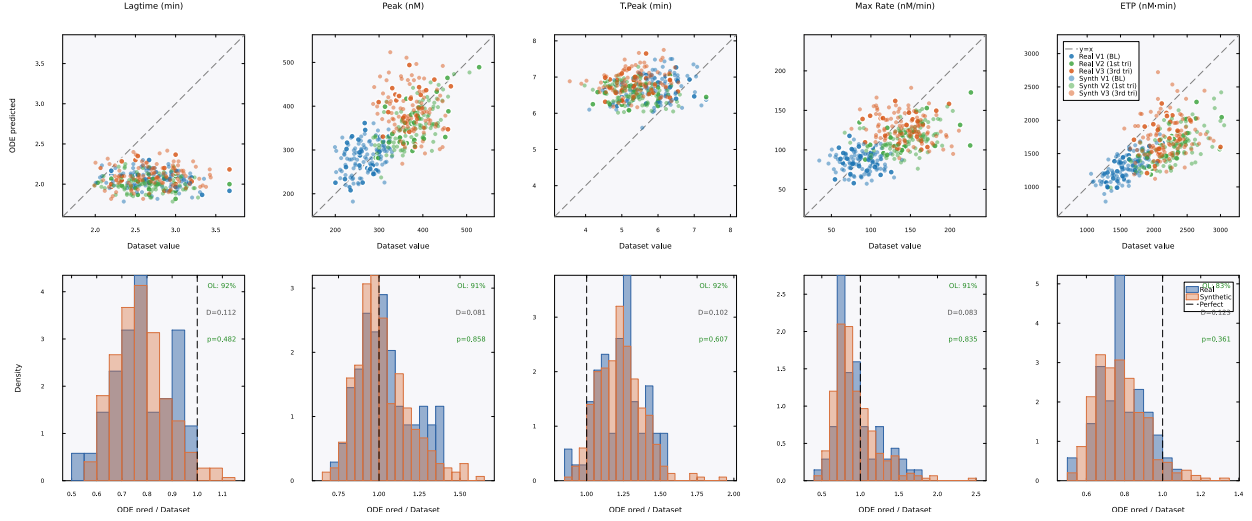


Figure 6: **Mechanistic validation (TF-only).** Top row: BZ2012 ODE-predicted TGA values (vertical axis) versus dataset TGA values (horizontal axis) for five thrombin generation parameters. The dataset values are the TGA measurements from each patient’s record, present for both real and synthetic patients. The ODE-predicted values are computed by running each patient’s coagulation factor levels through the 58-species BZ2012 model. The dashed line indicates where the ODE model would perfectly reproduce the dataset value ($y=x$); systematic deviations from this line (e.g., lagtime underprediction) reflect ODE model bias, not synthetic data failure. Filled markers are real patients, open markers are SA-generated synthetic patients, colored by visit: blue (V1/baseline, training set), green (V2/first trimester), orange (V3/third trimester). Real and synthetic patients occupy the same cloud with the same pattern of ODE model bias. Bottom row: ODE-predicted/dataset ratio distributions for real (blue) and synthetic (orange) patients. If SA had generated biologically implausible factor combinations, the ODE model would process them differently, producing divergent ratio distributions. Instead, the distributions overlap substantially. Each panel is annotated with cloud overlap (OL), the two-sample Kolmogorov–Smirnov statistic (D), and p -value. All five features show $p > 0.35$, confirming that the ODE model cannot distinguish real from synthetic patients.

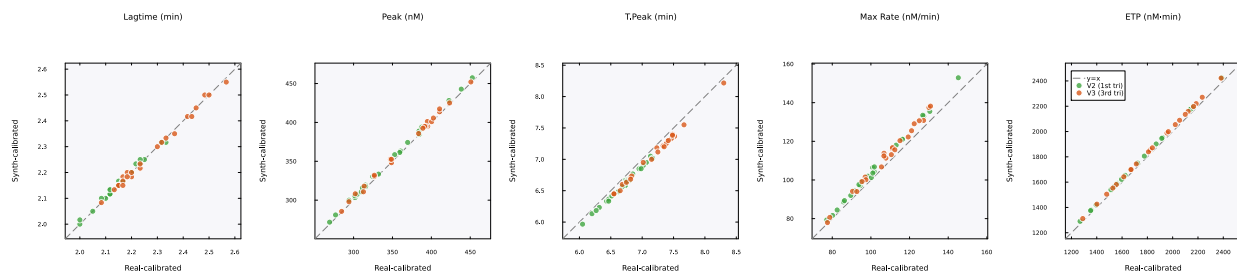


Figure 7: **Downstream utility: synth-calibrated vs. real-calibrated mechanistic model predictions on held-out real patients.** Each panel compares the BZ2012 ODE predictions for one TGA feature when the model is calibrated on real V1 patients (horizontal axis) versus synthetic V1 patients (vertical axis), evaluated on the same held-out real V2 and V3 patients. Points near the $y=x$ line indicate that the two calibrations produce equivalent predictions. Green: V2 (first trimester); orange: V3 (third trimester). The synth-calibrated model matched or outperformed the real-calibrated model on all five features (overall ratio $0.94\times$), demonstrating that a mechanistic model trained entirely on synthetic data generalizes to real patients as well as one trained on real data. Both calibrations used 11 random restarts; all starts converged.

Supplementary Information

Supplementary Tables

Table S1: **Inverse temperature sensitivity analysis.** SA generation quality as a function of β/β^* , where $\beta^* \approx 2.94$ was identified via entropy inflection in the 18-dimensional PCA space ($K=23$ stored patterns). Operating at $\beta = \beta^*$ (bold row) balances generation novelty against fidelity. Med MRE = median relative error of per-feature means; Novelty = $1 - \max_k \cos(\hat{\xi}, \hat{m}_k)$.

β/β^*	β	Noise scale	PC1 std ratio	Novelty	Med MRE (%)
0.1	0.29	0.261	0.538	0.542	0.7
0.3	0.88	0.151	0.550	0.536	0.7
0.5	1.47	0.117	0.562	0.528	0.7
1.0	2.94	0.082	0.589	0.502	0.6
1.5	4.41	0.067	0.613	0.474	0.7
2.0	5.88	0.058	0.638	0.446	0.7
3.0	8.82	0.048	0.689	0.403	0.8

Table S2: **Multiplicity weighting parameters for conditional generation.** The multiplicity ratio ρ was computed to achieve a target effective fraction $f_{\text{target}} = 0.80$ of softmax attention on the designated patterns. K_{eff} is the participation ratio quantifying the effective number of patterns contributing to generation.

Condition	K_{sub}	K_{bg}	ρ	f_{eff}	K_{eff}
Uncomplicated	18	5	1.11	0.80	23.0
PCOS	3	20	26.67	0.80	4.6
Developed PE	5	18	14.40	0.80	7.7

Supplementary Figures

Table S3: **SA generation hyperparameters.** The same values were used for unconditioned and conditioned generation except where noted.

Parameter	Value	Description
α	0.01	Langevin step size (ULA valid without MH correction)
T	2,000	Langevin iterations per sample
β^*	2.94	Inverse temperature (entropy inflection)
PCA threshold	95%	Variance retained
d_{PCA}	18	PCA dimensionality
N	100	Synthetic patients per cohort
Random seed	42	For reproducibility
f_{target}	0.80	Target attention fraction (conditioned only)

Table S4: **BZ2012 ODE model calibration.** Five rate constants were calibrated on Visit 1 real patients ($n=23$) using bounded Nelder–Mead optimization in log-scale-factor space. The cost function was the mean normalized relative squared error across five TGA features under TF-only conditions. Bounds: $\pm \log(100)$. Convergence: $f_{\text{tol}} = 10^{-6}$, 500 iterations, 11 random restarts.

Rate constant	Role	Default	Calibrated	Scale
prothrombinase k_{cat}	Thrombin burst	63.5 s^{-1}	21.94 s^{-1}	$0.35\times$
intrinsic Xase k_{cat}	Amplification Xa	8.2 s^{-1}	0.169 s^{-1}	$0.021\times$
extrinsic Xase k_{cat}	Initiation Xa	6.0 s^{-1}	93.2 s^{-1}	$15.5\times$
PC activation k_{cat}	Protein C activation	0.41 s^{-1}	0.890 s^{-1}	$2.17\times$
mIIa conversion k	mIIa $\rightarrow$ IIa	2.3×10^8	1.15×10^9	$5.01\times$

Units for mIIa conversion: $\text{M}^{-1}\text{s}^{-1}$

Table S5: **Summary of mean relative errors across all 72 features.** Per-visit distribution of MREs between real and SA-generated synthetic patient means. The full feature-by-feature table ($72 \text{ features} \times 3 \text{ visits} = 216 \text{ entries}$) is provided in the code repository as `paper_summary_statistics.csv`.

Visit	Median MRE	Mean MRE	25th pctl	75th pctl	Max MRE
V1 (baseline)	0.019	0.027	0.010	0.033	0.157
V2 (1st tri)	0.019	0.027	0.009	0.034	0.124
V3 (3rd tri)	0.016	0.025	0.006	0.040	0.110

Table S6: **Comparison with deep generative baselines.** CTGAN and TVAE were trained on the same 90 per-visit records (23 patients $\times$ 3 visits, 72 features) at three epoch counts. SA and MVN results from the main text are shown for reference. CTGAN fails at this sample size (median MRE $\approx$ 19%), consistent with GAN mode collapse on small datasets. TVAE achieves comparable marginal fidelity at high epoch counts but cannot capture cross-visit covariance or perform conditional generation.

Method	Epochs	Med MRE	Med KS	Corr MAE
SA	–	0.019	0.169	0.200
MVN	–	0.024	0.161	0.090
CTGAN	300	0.189	0.364	0.261
CTGAN	1,000	0.195	0.349	0.265
CTGAN	3,000	0.187	0.341	0.180
TVAE	300	0.037	0.174	0.152
TVAE	1,000	0.026	0.248	0.082
TVAE	3,000	0.018	0.224	0.092

Table S7: **PCA variance threshold sensitivity analysis.** SA generation quality is stable across thresholds from 85% to 99%: median MRE remains between 1.2% and 1.8%. The 95% threshold used in the main text (bold) is not a critical design choice.

Threshold	d_{PCA}	K/d	β^*	Novelty	Med MRE	Corr MAE
85%	13	1.77	2.94	0.413	0.012	0.116
90%	15	1.53	2.94	0.462	0.018	0.131
95%	18	1.28	2.94	0.516	0.012	0.148
98%	20	1.15	2.94	0.538	0.013	0.127
99%	21	1.10	2.94	0.540	0.013	0.133

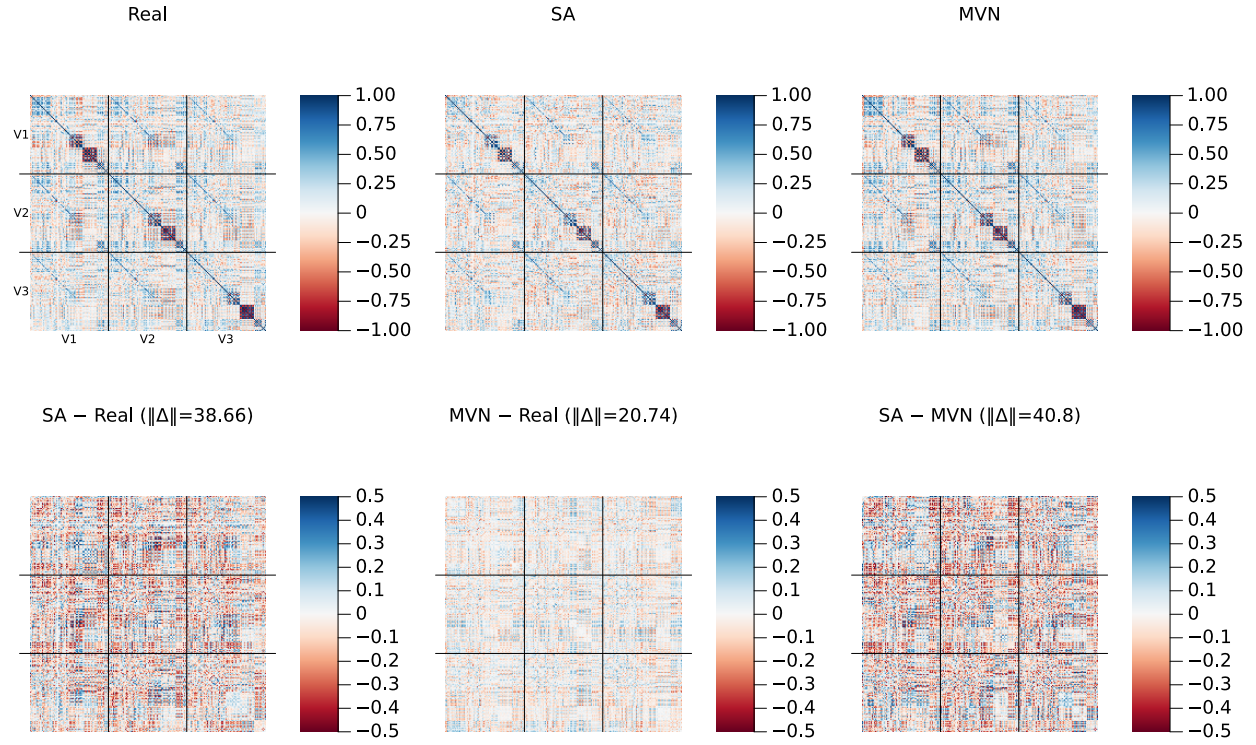


Figure S1: **Cross-visit correlation structure (amplified residual scale)**. Same layout as the main-text correlation figure, but with the bottom-row residual panels plotted on a ± 0.5 color scale (vs. ± 1.0 in the main text) to reveal fine structure. Frobenius norms of each residual matrix are shown in the panel titles.

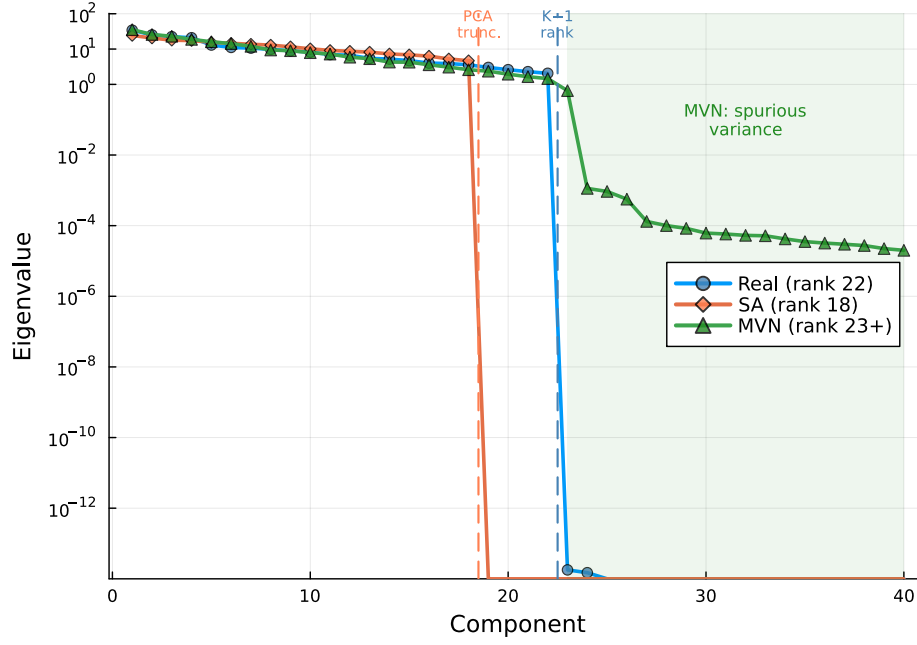


Figure S2: **Covariance eigenvalue spectrum.** Log-scale eigenvalues of the 216×216 sample covariance for Real ($K=23$, rank 22), SA ($N=100$, rank 18), and MVN ($N=100$) populations. Dashed vertical lines mark the SA PCA truncation point (component 18) and the data rank limit (component 22). MVN’s Ledoit–Wolf regularization inflates eigenvalues beyond component 22 (green shaded region), introducing spurious variance in 194 dimensions where the data contain no signal.

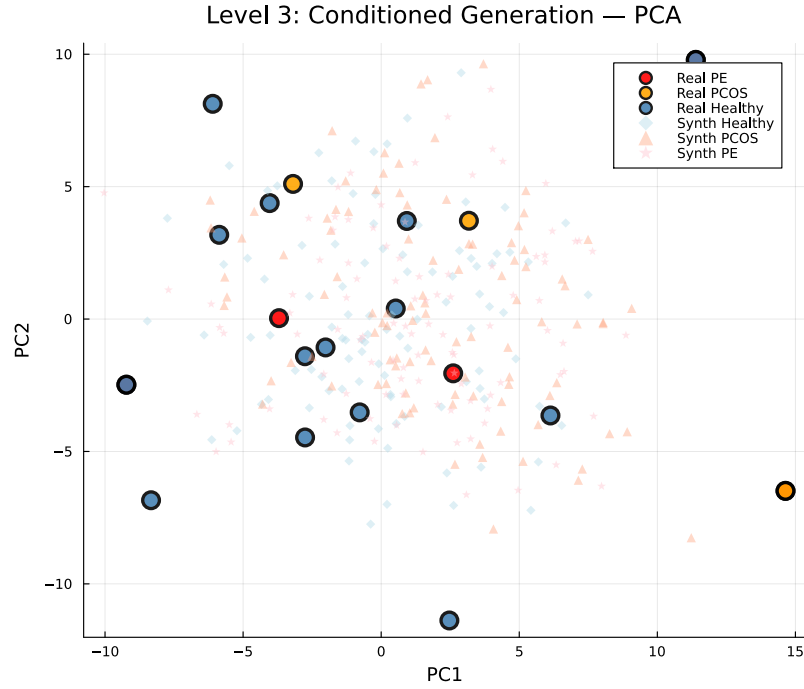


Figure S3: **Conditioned generation: PCA projection (pooled)**. Large markers: real patients (blue = Uncomplicated, $n=18$; orange = PCOS, $n=3$; red = Developed PE, $n=5$). Small markers: synthetic patients ($N=100$ per condition). With only 3 PCOS and 5 Developed PE patients, the subgroups do not form clearly separable clusters in the first two principal components, but the synthetic cohorts concentrate around their respective real counterparts.

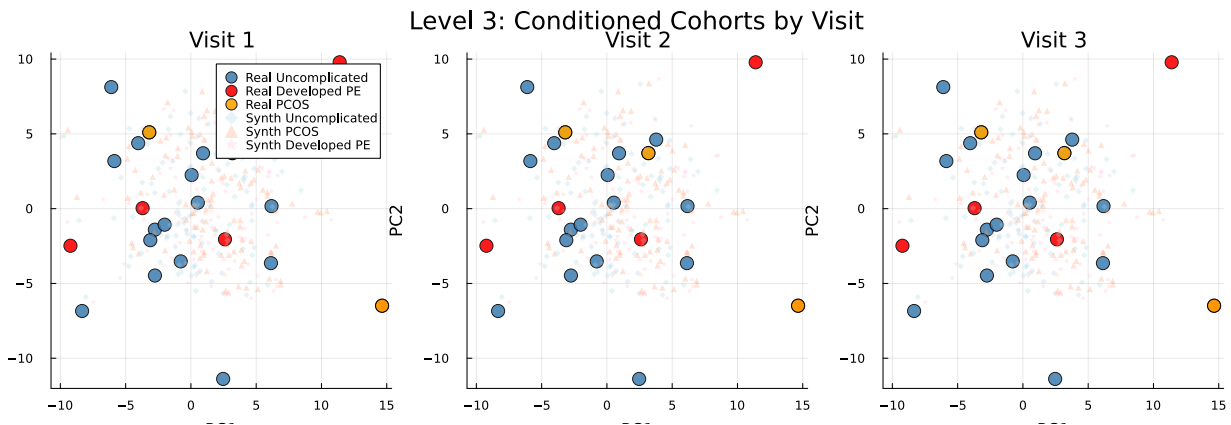


Figure S4: **Conditioned generation: PCA projection by visit**. Same as Supplementary Fig. S3 but separated by visit (V1, V2, V3). Condition-specific clustering is maintained within each individual visit.

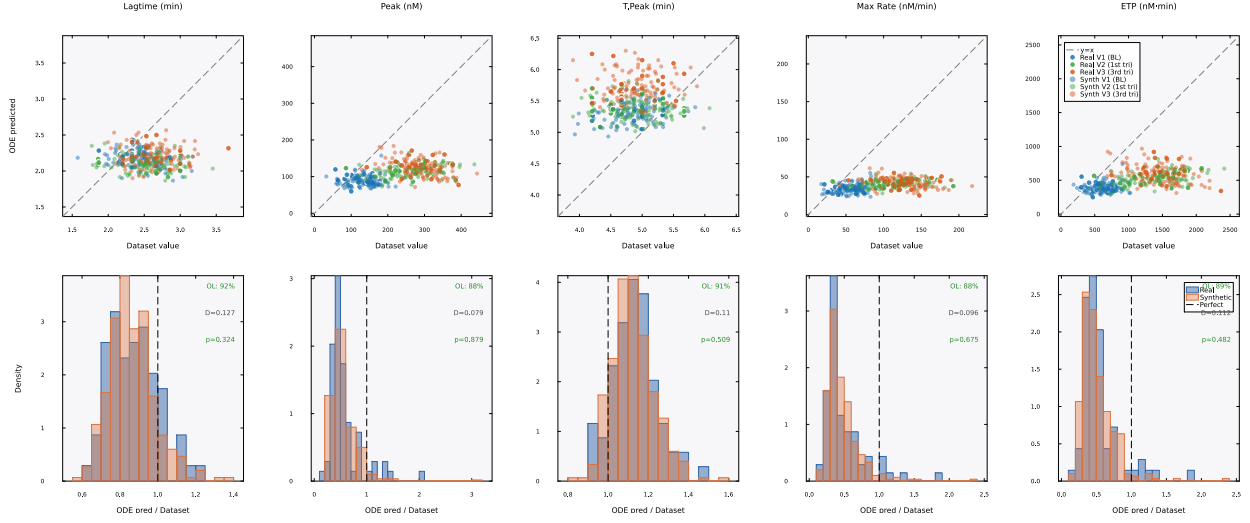


Figure S5: **Mechanistic validation (TF+TM condition).** Same layout as main-text Fig. 6 but under TF+TM conditions. The model systematically underpredicts peak and maximum rate ($\approx 0.5\times$ measured) for both real and synthetic patients, reflecting overcorrection of the protein C anticoagulant pathway. Cloud overlap remains high at 88–92%, confirming that the model processes real and synthetic patients identically even under imperfect calibration conditions.

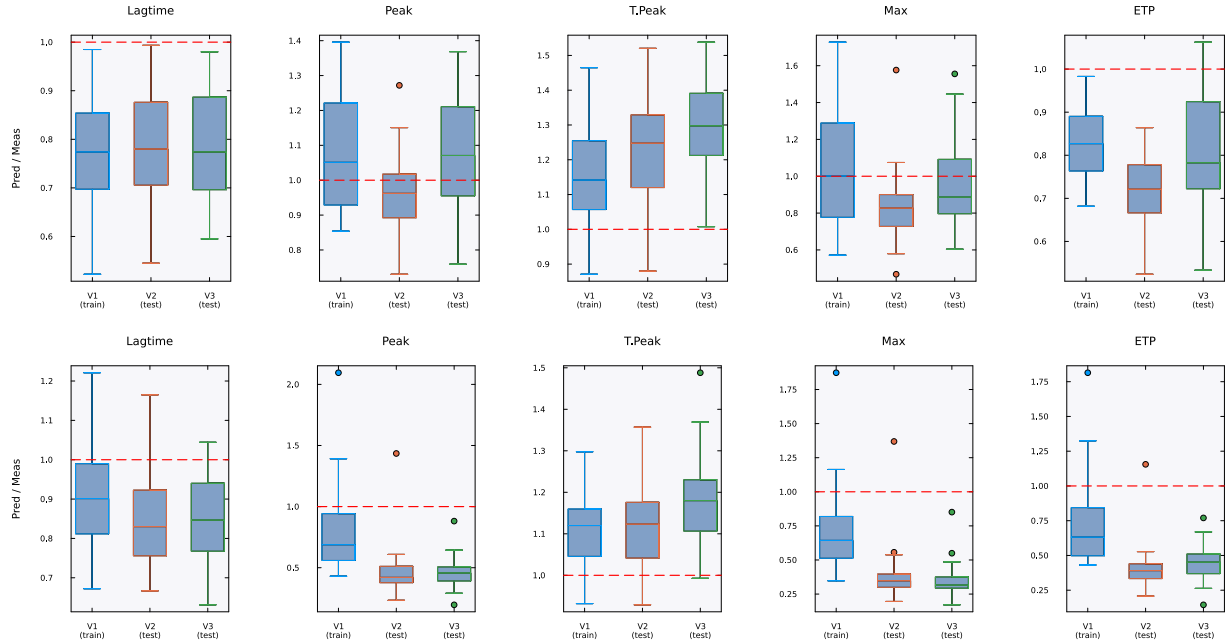


Figure S6: **Cross-visit calibration generalization.** Boxplots of the ODE-predicted/dataset ratio by visit for real patients. The model was calibrated on Visit 1 only. Top: TF-only conditions, where ratios near 1.0 at Visits 2 and 3 indicate that the calibration generalizes across pregnancy timepoints. Bottom: TF+TM conditions, where the calibration does not generalize as well, particularly for peak and maximum rate.

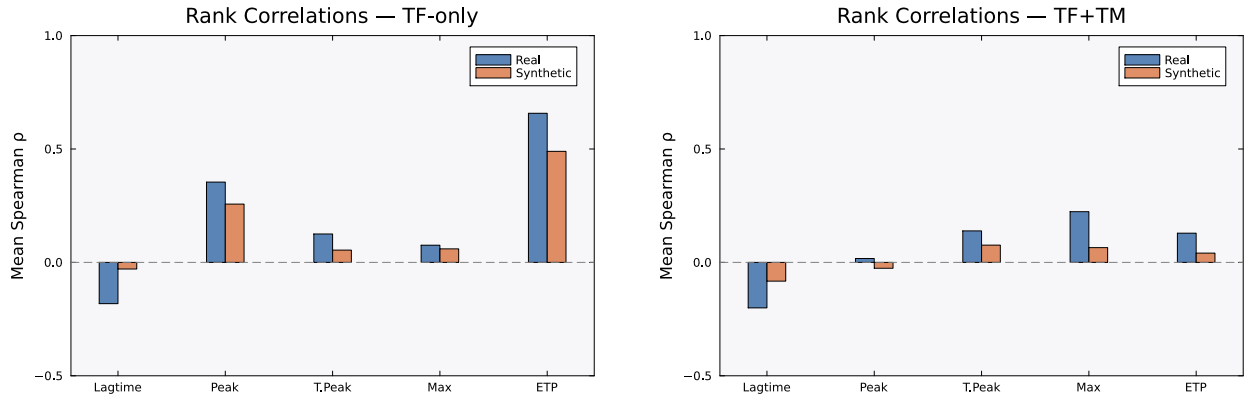


Figure S7: **Spearman rank correlations between ODE-predicted and dataset TGA features.** Left: TF-only. Right: TF+TM. ETP is the best-predicted feature ($\rho \approx 0.6$ – 0.8 for real patients under TF-only). Weak rank correlations for other features are expected given the 5-parameter population-level calibration.